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ABSTRACT

Vision-Voice: Smart Assistant for Visually Impaired is a cutting-edge mobile and web-based application designed to promote the independence and security of visually impaired people. This multi-purpose tool provides a range of features designed to help users in various settings. For indoor navigation, users read QR codes placed on doors or walls, enabling the app to offer detailed room information through Text-to-Speech (TTS). This facilitates smooth movement within buildings. For outdoor navigation, the app is integrated with Google Maps. Users just say their destination, and the app starts navigation while giving real-time directions using TTS, enabling hands-free navigation.

In case of emergencies, the application has an SOS alert feature that sends the user's current location using Google Maps and a distress message to a pre-configured contact with a single click of a button. The web application also incorporates a YOLO-based object detection system, which detects objects around and warns the user verbally so that they can avoid them and walk safely. Its second major feature includes a barcode-scan system which helps users inside supermarkets. If a user scans the barcode on a product using the app, it fetches the product details from a database of foods and notifies the user via TTS, helping with informed purchasing.

This project integrates state-of-the-art AI technologies, such as object detection and machine learning, with accessibility options, rendering daily activities safer and more convenient for the blind. Vision-Voice seeks to be a holistic and user-friendly assistant, offering dependable navigation, object identification, and emergency assistance in indoor and outdoor environments.

Keywords: Vision-voice, smart assistant, Visually impaired, Text-to-speech(TTS),Indoor & outdoor navigation, Object detection, SOS Alert, Barcode scanning, object identification, location sharing, YOLO-based object detection.

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LIST OF ABBREVIATIONS

TTS - Text-to-Speech.

QR - Quick Response.

YOLO - You Only Look Once.

8 **API -** Application Programming Interface.

GPS - Global Positioning System.

20 **ML -** Machine Learning.

CPU - Central Processing Unit.

GPU - Graphics Processing Unit.

ML Kit - Mobile Machine Learning Kit.

SOS- Save Our Souls.

1.INTRODUCTION

1.1 Motivation

The Vision-Voice: Smart Assistant for Visually Impaired initiative is driven by the wish of visually impaired individuals to lead more independent, safe, and convenient lives. According to WHO, it is reported that almost 2.2 billion people worldwide suffer from some kind of vision impairment or blindness. Most of these individuals face rather severe difficulties in doing very simple things such as navigating through public and private space, identifying objects within their environment, or in some instances much more fundamental things, such as shopping. There are no accessible technological solutions specifically aimed at these needs, and thus there is a significant barrier to their independence.

The motivation behind Vision Voice is the awareness that technological progress in AI, ML, and mobile apps can significantly fill this gap. The intersection of these technologies would be beneficial to develop a comprehensive tool for solving a number of the key challenges facing the blind-be it indoors, outdoors, or in some specific context such as shopping or emergencies.

The problem of indoor navigation still remains a challenge to many visually impaired individuals or GPS is not everyone outside. QR codes have turned out to be the most affordable and easiest method of indoor navigation. Google maps and voice instructions ensure smooth and audited movement for the blind in public places. In addition, the SOS feature responds to individual security concerns by delivering people an instant exit strategy to seek assistance during emergencies.

It has created an object detection functionality that would avoid visually impaired individuals from bumping into objects and enable them to move around freely, enhancing their mobility and confidence simultaneously. Importantly, it has integrated the shopping guide function, where a visually impaired individual would be able to shop alone by reading the barcodes of items to get information on each item from the product database.

Overall, Vision-Voice is brought about by the vision to have a world where technology provides the gate to inclusion and accessibility for everyone. It strives to enhance the quality of life for visually impaired individuals by equipping them with devices that construct their self-enabling and security in varying environments.

1.2 Problem Statement

Tactically, the problem that is being addressed by the Vision-Voice: Smart Assistant for the Visually Impaired project is the non existent accessible holistic technology solution that would help a visually impaired individual engage in different environments or carry out some basic activities independently. An estimated over 2.2 billion people in the world live with vision impairment or blindness, as per the World Health Organization, with every person encountering a multitude of barriers within various realms of life like being indoors or outdoors, recognition of objects, safe existence for personal well-being, and common daily activities such as going for a purchase.

GPS-based passive navigation systems would not provide support for indoor settings, thereby making visually impaired individuals reliant on moving individuals for mobility with new areas. Obstacle avoidance and object detection within the environmental context continue to be significant issues as most mobility aids and guide dogs fail to give timely feedback or even sufficient information. Outdoor navigation is partially by-the-fay facilitated through GPS-based access software.

Personal safety has emerged as another extremely important factor, with emergencies necessitating visually disabled persons to notify others and convey their locations. Also, shopping independently is quite a task, since the visually disabled have it very hard to distinguish between products and get relevant information (e.g., names or prices) on them, and distinguish between identical products on shelf space in shops.

Vision-Voice project intends to address these challenges by offering a multi-purpose smart assistant featuring indoor and outdoor navigation, real-time object recognition, an SOS emergency feature, and barcode reading to facilitate shopping. The suggested solution applies artificial intelligence and voice technology to enable the visually impaired to have freedom, safety, and independence to live on their own.

1.3 Project Objectives

The main goal of the Vision-Voice: Smart Assistant for Visually Impaired project is to improve the independence, mobility, and safety of visually impaired people through a holistic and easy-to-use mobile application. The project seeks to utilize advanced technologies such as artificial intelligence, computer vision, and voice interaction to develop a multi-purpose tool that solves major problems encountered by visually impaired people in their daily activities.

The key objectives of this project are as follows:

1.Indoor Navigation Support: To offer a stable and accessible means for visually impaired people to navigate indoor environments. Through the use of QR code scanning technology, the app enables users to scan codes positioned on doors or walls and receive in-depth room information via text-to-speech (TTS). This allows them to independently navigate unfamiliar places such as offices, malls, or hospitals.

2.Outdoor Navigation Assistance: To enable safe and efficient outdoor mobility with the aid of voice inputs and Google Maps integration. Through voice-enabled input for destination choice and real-time guidance via TTS, the app is designed to assist visually impaired individuals on the go through step-by-step directions that enable them to move about with confidence.

3.Emergency SOS Feature: In order to enhance personal safety by providing an emergency alert system. By a single touch, users are able to send their current location and an emergency message to pre-programmed contacts, providing prompt help in situations of urgency.

4.Real-Time Object Detection: To enhance mobility by integrating a YOLO-based object detection system that scans the environment and informs users of nearby objects through TTS. This helps users avoid obstacles and navigate their surroundings more safely.

5.Shopping Assistance: For helping blind individuals in supermarkets by providing them with barcode scanner and product recognition functionalities. By utilizing a food database, the application gives users useful information about products to make correct buying decisions.

1.4 Existing Systems:

Over the past few decades, multiple technological solutions have emerged to aid visually impaired people in understanding their environment, detecting objects, and executing everyday activities. The current works and approaches give significant information on how to create intelligent assistants such as Vision-Voice. The assistive technologies for visually impaired people in literature cover multiple domains, including navigation, object detection technology, and personal safety devices.

1. Indoor Navigation: Navigation indoors for the blind has proven to be a difficult issue because there is no GPS signal indoors. Beacon-based systems and QR code navigation have been considered to solve this problem. Bluetooth technology-based beacon-based systems provide location-based directions, but they need substantial changes in infrastructure. QR code-based solutions, as applied in Vision-Voice, are less expensive and easier to use. Research indicates that QR codes, when scanned with mobile phones, have the ability to provide individual room and location information, proving a viable option for indoor navigation.

2. Outdoor Navigation: Outdoor navigation aids for the visually impaired usually use GPS and speech interfaces. Most applications, for example, Google Maps, already have features such as turn-by-turn guidance and voice instructions. Visually impaired users, however, need more support, including real-time feedback and audio alerts to facilitate safe and effective navigation. Studies have investigated the integration of GPS with speech recognition and text-to-speech (TTS) technology to enable users to input commands via voice and receive real-time directions.

3. Object Detection: Object detection has been greatly enhanced with the of deep learning models such as YOLO (You Only Look Once) and OpenCV for real-time image processing. These models enable precise object detection in the environment, enabling visually impaired people to navigate around obstacles. Current systems such as Aira and Seeing AI offer similar features, enabling users to obtain real-time information regarding objects in their environment through smartphone cameras or wearable technology. Vision-Voice extends these developments by combining YOLO for object detection with TTS feedback.

4. Shopping Assistance: For shopping, apps such as BeMyEyes and NaviLens assist visually impaired people by providing product identification through scanning of barcodes

or QR codes. These applications match users with sighted volunteers or query databases to find product information. Vision-Voice also incorporates barcode scanning, allowing users to scan products and obtain information through TTS, providing a more independent shopping experience.

1.5 Limitations of Existing Work

In spite of the huge progress made towards assistive technologies for the blind, current solutions still have various limitations. The limitations are such that they impact their efficiency, usability, and practicality in real-time situations. Some of the main limitations of the current systems are discussed below:

1.Indoor Navigation Challenges: Most of the current indoor navigation solutions, e.g., beacon-based or Wi-Fi triangulation solutions, involve widespread modification of the infrastructure. These are based on the deployment of beacons or other hardware in indoor spaces, which is expensive and time-consuming. In addition, these solutions are not always present in public areas like malls, hospitals, or offices. QR code solutions are less expensive but frequently are based on the accurate positioning of the codes, which may not always be reliable or available in actual environments. This variability decreases the dependability of QR code systems for indoor wayfinding.

2. Outdoor Navigation Limitations: Although programs like Google Maps, employing GPS, allow blind travellers to navigate with voice commands, they are limited in their ability make particular, detailed real-time changes for the specific needs of the blind. They do not generally include features notifying individuals of barriers and abrupt elevation changes, i.e., curbs and steps, which are critical to mobility safety. Additionally, GPS receivers are not necessarily accurate in densely populated city centers, which can lead to confusion among visually impaired people. Furthermore, in noisy outdoor conditions, voice feedback alone using TTS may be problematic, with ambient noise overpowering navigation directions.

3. Object Detection Shortcomings: While recent advances in deep learning algorithms such as YOLO and technologies such as OpenCV have facilitated real-time object detection, accuracy in these systems can be impaired by low lighting, high speeds, or occluded environments. Some current object detection applications, such as Seeing AI, simply offer minimal data regarding objects within proximity without returning contextual or proximity

information. This may hinder the user from easily evading objects or navigating difficult spaces.

4. Limited Autonomy in Shopping Assistance: Existing shopping guide applications like BeMyEyes or NaviLens usually involve the use of third-party volunteers or barcode databases, which might not always be current or complete. They also do not include functionality for visually impaired individuals to manage shopping lists or monitor purchases, restricting the entire shopping experience and autonomy.

1.6 Proposed System

The Vision-Voice application is an intelligent assistant designed for visually impaired users, giving them the tools they need for navigation, object recognition, emergency response, and product identification. The system addresses the shortcomings of current solutions by combining several assistive technologies into one, user-friendly application.

1. Indoor Navigation Using QR Codes: For indoor wayfinding, the system employs QR codes installed in strategic positions including doors and corridors. Upon a user scanning a QR code via their smartphone's camera, the system interprets the data saved within the QR code, such as information concerning the purpose of the room as well as relevant directions. It then translates such information into words via Text-to-Speech (TTS) technology, providing users with audible instructions regarding where they are. This characteristic allows a blind person to independently navigate intricately structured interior spaces like offices, schools, and hospitals.

2. Outdoor Navigation with Voice Commands and Google Maps: The outdoor navigation component takes advantage of Google Maps and voice recognition technology to guide the users to their locations. The users can enter their destination using voice instructions, avoiding manual text entry. The system interprets the speech input, acquires the location, and calculates the most efficient path through Google Maps. As the user moves around, real-time GPS tracking provides ongoing feedback of their progress. The system gives turn-by-turn audio directions using TTS so that the user is able to drive safely without the need for visual assistance.

3. Object Detection with YOLO and OpenCV: To improve spatial awareness, the system combines YOLOv8 , a robust object detection algorithm, with OpenCV for real-time image

processing. The smartphone camera takes a snapshot of the user's environment, and the system identifies objects in the view, like furniture, doors, or obstacles on the path. Announced through TTS, the identified objects give users immediate feedback regarding their surroundings. This function makes movement safer and avoids unintended collisions.

4. SOS Emergency Alert System: In the case of emergencies, the system contains an SOS alarm module in which users can request assistance with a single click of a button. Upon activation, the system captures the GPS coordinates of the user and sends out an emergency message as well as a Google Maps link to their pre-specified contacts or emergency services automatically. This rapid-response facility ensures that visually impaired persons are able to get instant help in emergency situations.

5. Product Identification with Barcode Scanning: The app also features a barcode scanning mode to support users when they are shopping. By employing the camera on their smartphone, users can scan the barcode of a product, and the system extracts relevant information from an associated database. The product name and description are then read out by TTS to enable users to recognize items without needing visual verification. This capability improves their independence when buying commodities from supermarkets or shopping malls.

1.7 Advantages of Proposed System

1. Enhanced Indoor Navigation

One of the primary advantages of the system is its ability to facilitate indoor navigation using QR code scanning. Unlike traditional systems that rely on complex and costly infrastructure, this solution is cost-effective and easy to implement. Users can quickly scan QR codes placed at different locations, and the system will provide detailed audio descriptions, enabling independent movement inside buildings such as malls, offices, and hospitals.

2. Voice-Assisted Outdoor Navigation

The system takes advantage of Google Maps and voice recognition for flawless outdoor guidance. Rather than typing destinations manually, users just have to state their desired place, and the system will plan the shortest path. Real-time GPS navigation makes the

system's guidance precise, and Text-to-Speech technology gives audible instruction for navigation, making it very convenient for people who are visually impaired.

3. Real-Time Object Detection for Safer Navigation

Using YOLOv8 and OpenCV, the system is capable of detecting objects in real-time and alerting users to obstructions in their way. This minimizes the likelihood of accidents and injuries since users are constantly aware of what is around them. In comparison to conventional white cane or guide dog-based travel, this solution offers a virtual layer of safety without the need for further external aid.

4. Quick and Reliable Emergency Assistance

The SOS alert functionality enables users to request assistance by simply pressing a single button. The system automatically dispatches an emergency message along with the user's GPS position to pre-designated contacts or emergency services. This provides prompt response in critical situations, enhancing the safety of visually impaired travellers when they travel alone.

5. Independent Shopping Experience with Barcode Scanning

The barcode scanning capability allows users to scan products in retail stores or supermarkets independently without support. When a barcode is scanned, the system looks up product information from an attached database and speaks out the information through TTS. This facilitates the visually impaired to shop independently, boosting their confidence and independence.

6. Cost-Effective and Scalable Solution

Unlike costly wearable gadgets or sophisticated infrastructure-based solutions, Vision-Voice exploits commonly used smartphone technology. Using smartphone cameras, voice recognition, and machine learning models, the system delivers a cost-effective and scalable solution that can be implemented worldwide with little investment in new hardware.

2.AIM AND OBJECTIVES

2.1 AIM of Proposed System

The Vision-Voice application is intended to act as an intelligent assistive system for the visually impaired, allowing them to move around safely and independently. The main objective of the suggested system is to incorporate state-of-the-art computer vision, speech processing, and navigation technology in order to offer users a seamless experience in both outdoor and indoor settings. By utilizing QR codes, Google Maps, YOLO-based object detection, an SOS alert system, and barcode scanning, the system provides a holistic solution for visually impaired people, covering their mobility and accessibility issues.

Perhaps most saliently, one of the system's most visible objectives is the provision of accurate and real-time navigation instructions. The indoor navigation technology achieves this by utilizing QR codes to enable users to identify rooms and spaces in buildings and move around freely with no external support. Similarly, the outdoor navigation component is designed to be accessible via voice guidance and Google Maps, so users can type in destinations vocally and receive real-time navigation instructions via Text-to-Speech (TTS). This eliminates the need for manual input and makes it more convenient.

Again, another key goal of the system is to improve spatial awareness and obstacle detection. YOLOv8 and OpenCV integration enables real-time object detection such that users are provided with audio notifications on objects ahead of them, thereby reducing accidents. Additionally, the SOS emergency alert system enables a safe mode through sending live location updates to emergency contacts, thereby allowing users to access quick rescue when necessary.

In addition, the system will ensure independence in daily tasks, including shopping, via barcode reading. Through the retrieval and verbal reading of product information, blind people can independently make purchases without any assistance.

2.2 Objectives of Proposed System

1. To Provide Seamless Indoor Navigation

21 One of the most important objectives of the proposed system is to enable blind individuals to navigate within indoor spaces efficiently and safely. Using QR code navigation, the system grants users the power to scan the QR codes placed at different positions throughout the area, such as doors or corridors, and receive real-time Text-to-Speech (TTS) feedback regarding their location and surroundings. This minimizes reliance on external assistance and boosts autonomy in confines such as malls, hospitals, schools, and offices.

2. To Facilitate Voice-Assisted Outdoor Navigation

The system is designed to combine Google Maps and speech recognition technology for voice-enabled outdoor navigation. Rather than forcing users to enter their destinations, the system interprets verbal instructions, recognizes the location, and creates an optimized route. The driving instructions are delivered using TTS technology to provide users with constant and accurate directions throughout the journey.

3. To Enhance Obstacle Detection and Spatial Awareness

To achieve greater mobility and prevent accidents, the system applies YOLOv8 (You Only Look Once) and OpenCV to detect objects in real-time for object detection. Through instant auditory warnings from object and obstacle detection, the system allows users to navigate safely within their surroundings. This purpose is critical in minimizing the risk of collisions as well as situational awareness in total.

4. To Ensure Emergency Assistance with an SOS Alert System

Security is a built-in aspect of the proposed system. The SOS alarm module enables people to send emergency alarm with location via GPS at the press of a button to pre-registered individuals. The users are ensured to receive quick help in the case of emergency, accidents, or distress since the system can respond quickly, thereby reducing rescue times. The system is on a mission of reducing emergency rescue times and providing a safety network for visually disabled individuals traveling alone.

5. To Enable Independent Product Identification in Retail Environments

The proposed system has a barcode scanning feature that allows customers to scan barcodes of supermarket products and retrieve corresponding product information from a connected database. The information is read out by TTS, allowing visually impaired people to make a purchase decision on their own without the assistance of others. This objective enhances the user's autonomy in shopping environments.

6. To Develop a Cost-Effective and Scalable Solution

Another important goal is to make the system cost-effective, scalable, and deployable. In contrast to high-cost assistive devices, the system is built to operate using standard smartphone cameras and processing, which can reach a larger number of people without the need for specialized hardware.

3.LITERATURE SURVEY

3.1 Introduction

It has been decades since the development of various assistive devices for the blind, following the progress of computer vision, artificial intelligence, and speech processing. These devices have ranged from white canes and guide dogs, which have extremely short ranges, need physical contact with the object to be touched or marked, and do not have sophisticated contextual information about the environment in relation to the location. In recent years, mobile phone- or computer-based applications from artificial intelligence-capable models have provided potential solutions for improving navigation, object detection, and independent living for the visually impaired.

At first, assistive technology was all about detection of obstacles by audio or haptic feedback using navigation systems based on ultrasonic sensors. However, the performance of such systems is decreased extensively with very high restrictions on range and accuracy. Computer vision and deep learning-based methods like You Only Look Once (YOLO) and OpenCV later arrived with capabilities of real-time object detection, giving alarm warnings to users regarding obstacles on the way in the least amount of time.

Voice-guided navigation systems continued to improve based on Google Maps API and speech recognition, provided users with outdoor navigation. Primarily, such systems provide scope to the blind and offer a way through which they can issue voice instructions and get step-by-step directions with the help of a Text-to-Speech (TTS) engine. Another low-cost technology for indoor navigation is QR code scanning because of the previously mentioned methodology, in which the user can scan a QR code carrying information about the specific room and employ it while navigating in much larger indoor spaces.

Emergency response techniques were another key area of research that evolved SOS alert apps, with GPS tracking to enable the user to call emergency individuals with current location information. Barcode scanning of goods for identification was still incorporated into these support apps so that users can identify common products during shopping.

3.2 Existing Work

The assistive technology innovation for visually impaired individuals enables numerous tools for mobility, object identification, and greater interaction with the world. One example is Assistive Object Recognition System designed by Shaikh, Karale, and Tawde in 2020. The suggested approach is a YOLOv3-based object identification and detection system with Raspberry Pi as the interface to provide feedback. The present discovery shows a general range of performance between 85% and 95% but restricts the detection of 100% in certain objects such as individuals, chairs, clocks, and cell phones.

This system is the Smart Navigation System Help for Visually Impaired Individuals, courtesy of the National Institute of Electronics & Information Technology labs, otherwise referred to as NIELIT. The system suggested relies on an ESP32 microcontroller with radar and ultrasonic sensors so that it reports on the hindrances in the way. The data is translated into sound signals that would assist in navigation. The Bluetooth mechanism is an audio navigation feature that enhances the user interface and brings the device to user-friendliness, lightness, and power efficiency.

This system is intended to allow visually impaired persons to deduct oncoming vehicles at a distance. Computer vision techniques like segmentation, object detection, and optical character recognition (OCR) are employed in detecting and analysing the vehicle information, and this is communicated to the user. The VBRS has the benefit of being an accessible mobility keynote for all individuals within public transport systems.

Moreover, the VisiSense system offers a comprehensive IoT-based solution in terms of assistive technology. The aspect comprises a handstick with sensors, a viewing capture unit, and an edge-computing configuration for object detection and classification. VisiSense detects objects with 99% accuracy using some conditions based on deep learning methods such as R-CNN and MobileNet. Effective processing speed and resource consumption are showcased.

4. SOFTWARE REQUIREMENTS

4.1 Functional Requirements

There are functional requirements that the Vision Voice: Smart Assistant for the Blind project has. They are correctly defined and express the precise core functionalities and features introduced to the system to accomplish its main goals. Thus, they guarantee that the system enables the needs of blind users in real but realistic navigations, object identification, shopping assistance, and emergencies.

1. QR Code-based Indoor Navigation:

- The users must be empowered by the system to read QR codes on the doors or walls using a camera offered on their mobile devices. The QR code is then decoded and information about a room/location can be obtained.
- The engine must then read out the location information from the TTS engine for real-time output to the user. The QR codes must be scanned with reasonable precision within a range and also under varying light sources.

2. Outdoor Navigation with Voice Commands:

- The button, when pressed, should activate the Google Speech Recognition engine to receive input from a user vocally. The application should receive the destination from the user and initiate outdoor navigation with the Google Maps API.
- It must offer navigation directions to its users through turn-by-turn directions carried out with the TTS engine so that the user is provided with constant audio back support. The GPS tracking must remain true to its word and update and provide directions based on real-time user data.

3. Object Detection and Avoidance:

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- The application would use the mobile camera and the YOLO object detection model for detecting objects in real-time. The text-to-speech of objects detected shall be conveyed to the user to notify those of any obstacles that can be in their immediate surroundings.

- Object detection should perform effectively under different lighting conditions and different environments.

4. Shopping Assistance via Barcode Scanning:

- The mobile camera is able to scan product barcodes via the system. Once the scanning process is completed, the application fetches product information from the linked database and reads the information (i.e., product name, price, etc.) using TTS.

- The barcode readers must be of high accuracy for a large variety of products and must also be able to retrieve stored information without any failure.

5. Emergency SOS Alert:

- There should be an SOS button incorporated into the system such that, upon its activation, an emergency message along with the real location of the user is sent by SMS to specified numbers.

- The application should utilize the GPS module to get the current location of the user immediately and directly send the message.

4.2 Non-functional Requirements

Vision-Voice smart assistant must be designed with a performance and reliability focus in order to deliver effective and smooth performance to the visually impaired users. The system must respond in real time with a negligible lag, especially for applications that are time-critical in nature such as object recognition and SOS alert. The text-to-speech function must be activated very quickly so as to provide quick and clear audio response in order that users can provide instant feedback. The applications must have the capability of handling multiple requests with little latency, giving the user a seamless experience. The software must be able to scale and it must be giving proper information, with possibilities of future upgrade like AI-assisted support, broader recognition, and support for other languages for other citizens of the worlds.

Privacy and security are major features of the system, especially for functionalities involving GPS location tracking and user information. The app should offer end-to-end encryption to all communications, especially with the SOS alert feature, in order to provide the user with privacy. Safe login and logout processes such as PIN or fingerprint authentication should be done to prevent any unauthorized access. Storage and sensitive information such as locationhistory securely without misusing it.Data protection and access compliance, such as GDPR, must be maintained to maintain the rights of the users and privacy to see whether the users' sensitive information is being moved to third-party apps or not. Accessibility and usability of the app need to be planned well enough to accommodate visually impaired individuals.

The interface must be minimalist and voice-intensive, allowing users to access functions without visual interaction. Multi-language TTS capability must be activated by the system so that users can use the application in the language of their choice. The application must support all mobile screen readers and all the tools that are utilized for mobile phone accessibility. The application should also be able to perform optimally on various devices, even lower-cost smartphones, and a large number of people without necessarily needing costly devices.

4.3 Software Requirements

Vision-Voice: Smart Assistant for Visually Impaired project needs the whole software stack to be supported by its web and mobile facets. Android Studio is the primary integrated development environment (IDE) for developing Android applications, with the necessary tools for designing the UI, debugging, and runtime testing. Compilation and execution of the Android app need the Java Development Kit (JDK 8 or above). The web application created using Flask is efficiently using Python development tools like Visual Studio Code or PyCham for coding, debugging, and handling dependencies for flawless functioning. Since Flask is backed by Python 3.7 or later, the system should be configured in a proper way to make it compatible with libraries and frameworks as needed.

For outdoor navigation and object detection, YOLO (You Only Look Once), OpenCV, and deep learning frameworks such as TensorFlow or PyTorch are utilized to scan real-time video feeds and identify obstacles. The Android app utilizes the Google Maps API for outdoor navigation with turn-by-turn directions via Google Text-to-Speech (TTS) and

Google Speech API, which translate verbal commands into actionable text inputs. Besides, ZXing (Zebra Crossing) is also used for decoding barcodes and announcing the objects name, which helps visually impaired individuals identify goods in supermarkets and at home. The application requires SQLite or MySQL databases to keep product information and user data, and Flask-SQLAlchemy facilitates connections between the web application and databases.

External hardware devices also play a crucial role in system performance. QR codes must be printed and located at strategic points for indoor navigation, say in hospitals or shopping malls, while barcode scanning enables quick identification of products by users. Cloud-based navigation, speech recognition, and acquisition of data in real-time require a stable Wi-Fi or 4G/5G connection. The web app requires a 1080p webcam for image capture and object detection of high quality. If a flexible placement of external cameras is required, models such as the Logitech C920 offer higher resolution and low-light capabilities.

To collaborate and control versions, Git is used to track code change, and repositories at GitHub or GitLab are used for cloud storage and issue tracking. Android SDK and Gradle Build Tools render the system compatible with various Android versions, thereby facilitating an easy development and deployment process. Leverage this smooth technology stack, Vision-Voice efficiently brings mobility, autonomy, and access to visually impaired users through the use of AI, real-time processing, and cloud services.

4.4 Hardware Requirements:

The Vision-Voice: Smart Assistant visually impaired application is dependent on mobile and web-based modules like features of continuous object detection in real-time, QR code and barcode scanning via mobile app, navigation, and speech processing. For a successful operation, the system is in need of strong hardware and of course software modules, specifically to maintain tough processes like real-time image analysis and speech recognition. The application needs to be installed on a quad-core or above processor-powered Android smartphone capable of recognizing objects and scanning barcodes efficiently. The phone should have at least 2 GB RAM for trouble-free multi-tasking. The machine ought to possess at least 16 GB of internal storage capacity for the storage of maps and app data, and a back-facing camera with a resolution of at least 8 MP for scanning

barcodes and QR codes without interruptions. A-GPS hardware support is also required for navigation, and a satisfactory and ideal microphone and a good or improved quality speaker facilitate voice interaction through Google Text-to-Speech (TTS) and speech recognition APIs for smooth application execution for the app users.

For deployment and running of the project, the Flask-based web application is run and deployed on a hybrid or fast laptop/desktop machine with Intel i5 or AMD Ryzen 5 processor (or similar). At least 8 GB of RAM is required to test and create the web and mobile applications, though 16 GB or more is ideal to run machine learning models and object detection real-time processes. A 512 GB SSD is recommended for faster processing of data, particularly when using Android Studio, Flask, and object detection models like YOLO. An additional GPU (for example, NVIDIA GTX 1650 or higher) is also beneficial for accelerating image processing and deep learning object detection. When training the YOLO model locally, the GPU actually enhances performance with faster frame analysis and reduced latency.

External hardware devices also play a crucial role in system performance. QR codes must be printed and located at strategic points for indoor navigation, say in hospitals or shopping malls, while barcode scanning enables quick identification of products by users. Cloud-based navigation, speech recognition, and acquisition of data in real-time require a stable Wi-Fi or 4G/5G connection. The web app requires a 1080p webcam for image capture and object detection of high quality. If a flexible placement of external cameras is required, models such as the Logitech C920 offer higher resolution and low-light capabilities.

5. SOFTWARE DESIGN

5.1 Process Cycle

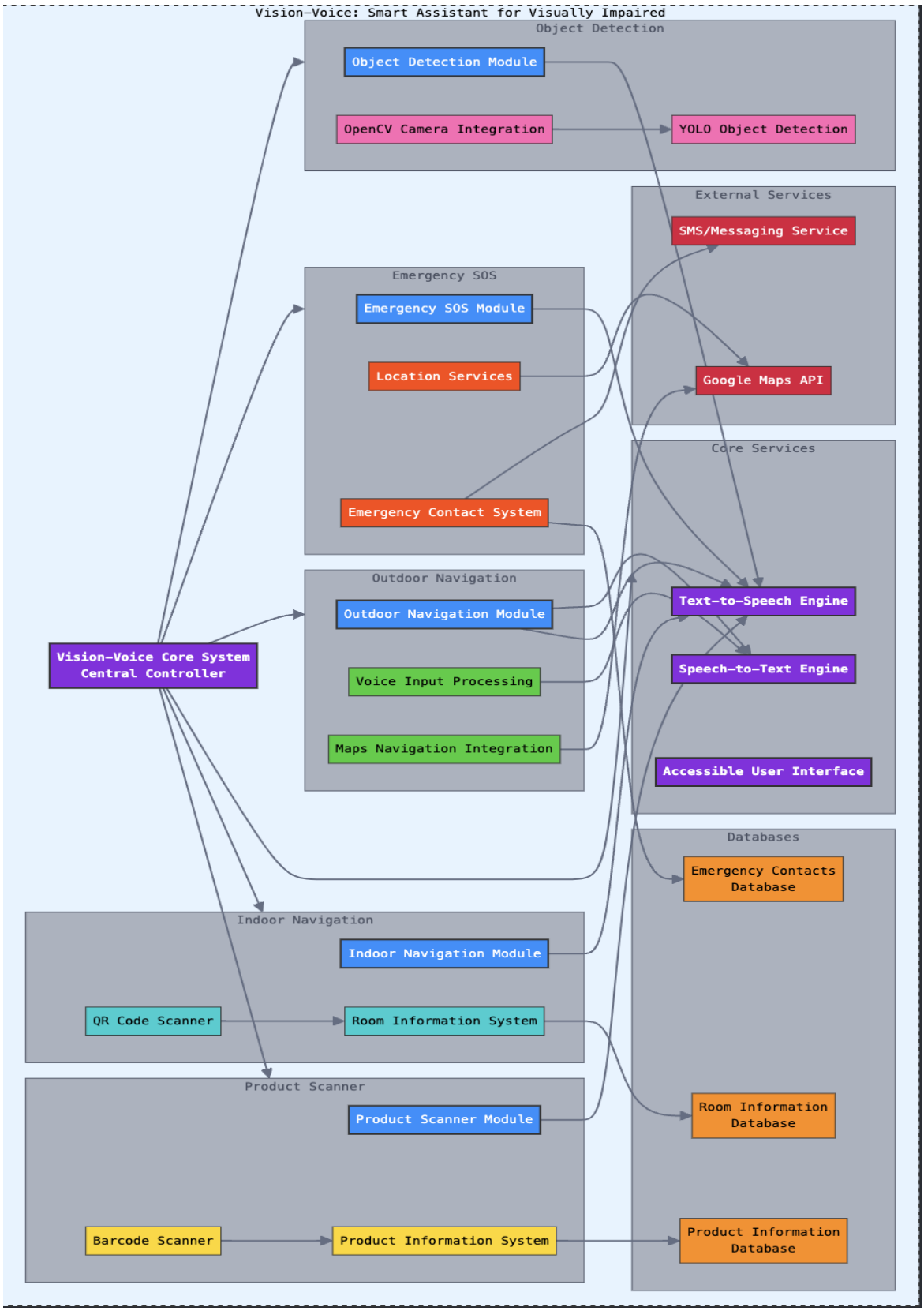

Fig 5.1: Architecture Diagram.

5.2 User Flow Diagram

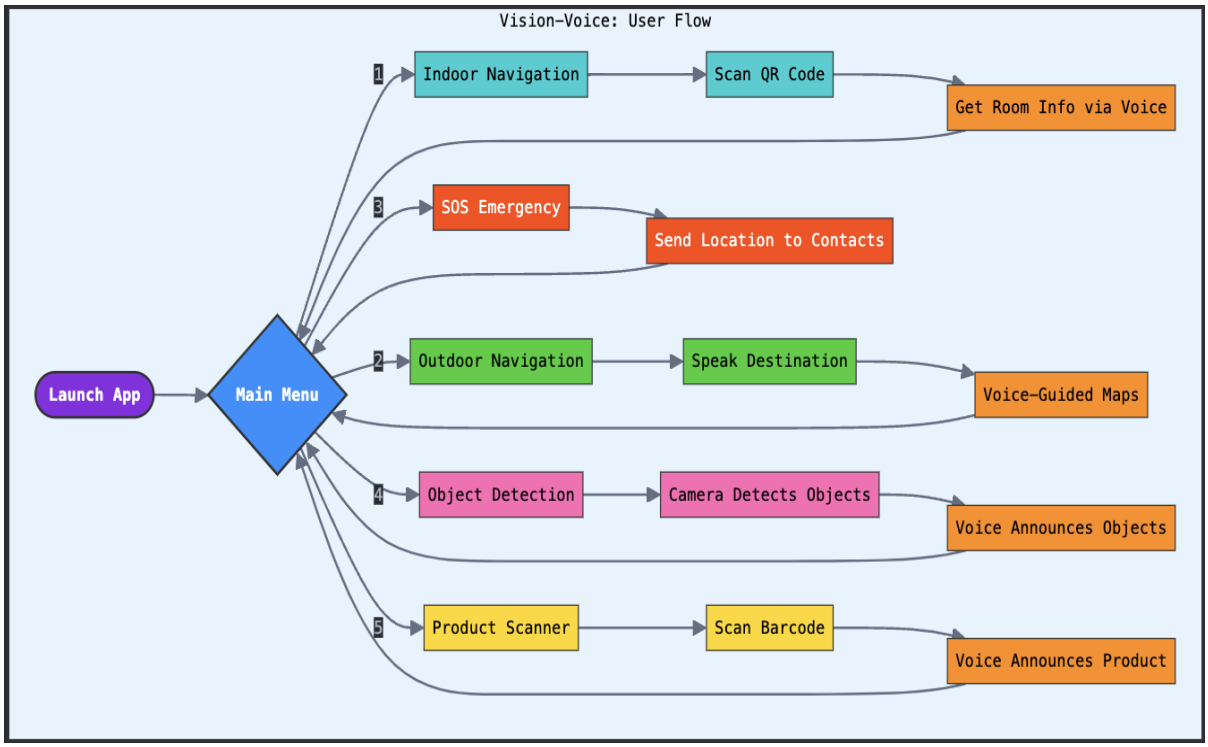

Fig 5.2: User Flow Diagram.

5.3 E-R Diagram

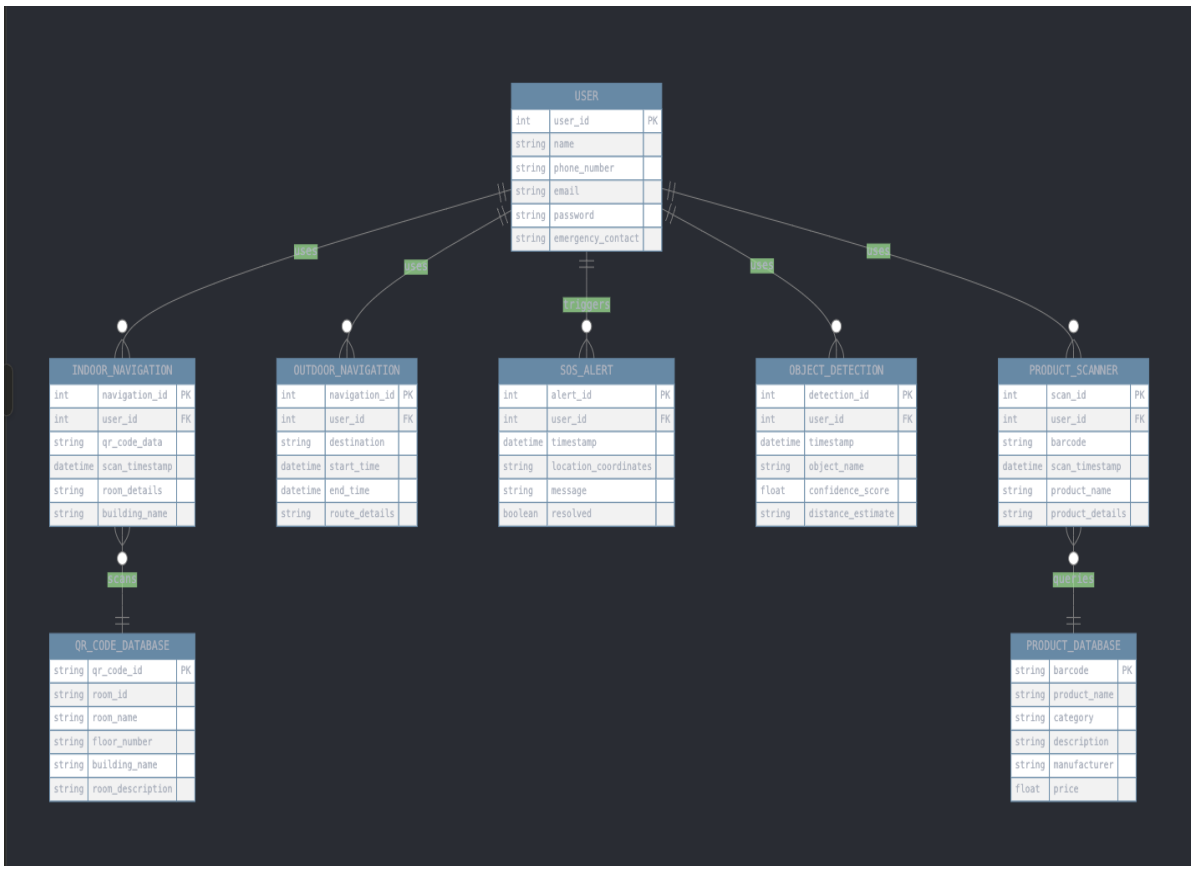

Fig 5.3: E-R Diagram.

5.4 Modules Description

Vision-Voice application comprises some major modules that are made for fulfilling various visually impaired users' requirements. They collaborate and enhance navigation, security, and autonomy in both indoors and outdoors.

1. **Indoor Navigation Module**: The outdoor navigation module is integrated with Google Maps and voice recognition. It allows users to input their destination through voice commands. Once the user provides a location, the app interfaces with Google Maps to calculate the best route. Using GPS data, the app tracks the user's location and provides real-time spoken directions through TTS, helping them reach their destination safely.
2. **Outdoor Navigation Module**: The outdoor navigation feature is also coupled with Google Maps and speech recognition. The feature enables users to enter their destination by voice. After the user gives a location, the application interfaces with Google Maps to determine the optimal route. With GPS information, the application follows the location of the user and offers real-time spoken instructions by TTS, enabling them to arrive at their destination safely.
3. **SOS Alert Module**: This module is intended to help users in times of emergency. With a single button press in the app, the SOS alert module takes the user's current GPS location and sends an emergency message, as well as a Google Maps link to the user's location, to pre-set contacts or emergency services. This functionality allows help to be quickly and easily requested.
4. **Object Detection Module**: With the use of YOLOv8 and OpenCV, the object detection module sweeps the user's environment for obstacles or objects. It identifies objects in real-time and informs the user of the object's name and location through TTS to provide safe navigation without collision.
5. **Barcode Scanning Module**: This module assists users in identifying products by reading barcodes on products. The application contacts a database of products, retrieves the product information, and speaks the information aloud, assisting visually impaired users in shopping by themselves.

6. SYSTEM IMPLEMENTATION

6.1 Indoor Navigation Implementation

The vision-voice project has an indoor navigation system aimed at helping visually impaired users navigate the inside of buildings using QR code technology and a Text-to-Speech (TTS) system. The process starts by positioning QR codes at the exterior of different rooms or spaces within a building. The QR codes hold information encoded in them pertaining to the identify of the room number, purpose, and other information like directions or markers.

When the user enters a room, they scan the QR code using their smartphone by pointing the phone's camera at the QR code through the app's built-in camera feature. The app makes use of the phone's camera to identify and read the QR code with a suitable QR code reading library (e.g., Google's ML Kit or Android's ZXing). When the QR code is identified, the system reads the encoded data and processes it.

he obtained data is further sent to the TTS engine, which turns the text information into speech. The TTS system (for example, Google TTS or another built-in engine) announces the room information to the user in real time. This helps the user locate where they are and get accurate information about the room or place they are heading into.

The deployment is made to make the experience very smooth. It accommodates accessibility features like voice instructions that direct the user to align the camera approximately and error handling for situations where the QR code is not scanned correctly. With this QR code-based indoor navigation system, visually impaired users are able to independently move around indoors without any external assistance, significantly enhancing their mobility and independence. The integration of real-time QR detection with TTS output provides a convenient and effective indoor navigation solution specifically designed for visually impaired users.

6.2 Outdoor Navigation and Google Maps Integration

The outdoor navigation technology in the Vision-Voice project offers a voice-guided, real-time navigation system to visually impaired people through voice commands, Google Maps, and Text-to-Speech technology. The process starts with an easy, user-friendly interface containing a large start navigation button. Upon activating the button, the system will initialize the Google TTS engine, asking the user for a destination.

The user then says the name of his or her desired destination, which is recorded by the system with the help of speech recognition software, like Google's Speech-to-Text API. The voice is translated into text and handled by the app. After the system identifies the location, it uses the Google Maps API to create a route from the current location of the user to the desired destination. This process entails asking Google Maps for proper directions, taking into consideration the user's location and any preferences (e.g., walking directions or public transport).

After the route is calculated, the app is in navigation mode, where Google Maps begins sending real-time instructions to the user. In this process, the app keeps on fetching the user's GPS coordinates to update the user's position on the map. With the TTS engine, the app translates the visual map directions into voice directions, instructing the user step-by-step with straightforward directions like "Turn left in 100 meters" or "Go straight for 200 meters."

The system is programmed to support visually impaired users with issues like regular users with issues like regular updates when they are about to take a turn or warning them if they need to reroute in the event that they stray from the intended route. In the event of straying or weak GPS signal, the system recalculates the route and continues to give updated directions through TTS.

In addition, the integration is designed to make the app deliver timely and accurate navigation feedback so that the user is able to navigate outdoor environments safely and confidently.By translating Google Maps' visual guidance into audio cues, the Vision-Voice outdoor navigation system allows visually impaired users to move about independently without needing extrinsic assistance. The integrated voice recognition, Google Maps, and TTS guarantee smooth and convenient outdoor navigation.

6.3 Object Detection and TTS Feedback

The object detection functionality in the Vision-Voice project aims to enable visually impaired individuals to safely move around the environment through the detection of objects in the vicinity and presenting real-time feedback through Text-to-Speech (TTS). The object detection function applies the YOLOv8 object detection model with OpenCV for the analysis of live camera feed from the smartphone's camera and the detection of objects in real time.

The deployment starts with accessing the camera stream via OpenCV, which feeds live video frames directly into the YOLOv8 model for object recognition. YOLOv8 is a fast and efficient object detection algorithm that can identify multiple objects with a single pass through the image. The model, which is pre-trained on COCO dataset, identifies a vast set of common objects like persons, chairs, doors, and vehicles, which are applicable for visually impaired users to navigate through different scenarios.

Every frame of the camera input is processed by YOLOv8, which identifies objects and provides bounding boxes, labels, and confidence per object. The identified objects are further processed to identify their location in relation to the user, i.e., whether they are on the left or right side of the screen. This spatial data is very important in aiding users to comprehend where the objects are in terms of their motion.

When the objects are identified, the system transmits the visual information as verbal feedback. The TTS engine pronounces the name of each identified object and offers spatial instructions, for instance, "Chair on the left" or "Person in front." The live audio feedback helps users make corrective movements and sidestep obstructions, which promotes safe mobility.

This object detection system is optimized to perform well even in dynamic environments, providing constant updates as the user moves around. The synergy between YOLOv8 for high-speed object detection and TTS for audio feedback presents a strong solution that improves the user's spatial awareness, allowing them to safely and independently navigate their environment.

6.4 SOS Alert System and Barcode Scanning

The SOS alert system and barcode scanning function in the Vision-Voice project are used to improve both the shopping experience and safety of visually impaired users. The SOS alert system is an important feature used to trigger assistance during times of emergency, while the barcode scanning function helps users identify products in settings such as supermarkets.

The SOS alert system is executed with an uncomplicated and easy-to-use interface—a special button in the app. Once a user taps the button during an emergency, the app proceeds to fetch the user's location immediately using GPS. The software produces a Google Maps link to the user's precise coordinates and sends this with a pre-programmed emergency message, e.g., "I am in an emergency," to a pre-programmed contact or emergency services as an SMS. The application of GPS guarantees that the location information is precise, while the auto-generated message obviates manual input during a traumatic experience. This characteristic is intended for speedy and trustful communication to make sure that assistance can be sent promptly.

The barcode scanning function benefits blind users by helping them identify products during shopping. It is very helpful in supermarkets where users should know the products they are selecting. The system utilizes the smartphone camera to scan product barcodes, just like QR code scanning is done for indoor positioning. The barcode information is matched against a product database to fetch corresponding information, e.g., the product name, price, and description.

Once the product details are retrieved, the application employs the Text-to-Speech (TTS) functionality to state out the product name and details in speech, enabling visually impaired users to make sound judgment while shopping. This functionality promotes greater independence for the visually impaired by avoiding visual verification of product information, hence providing a more independent shopping process. Both the SOS alert and barcode scanning systems operate in perfect harmony to enhance the safety, convenience, and independence of visually impaired users in day-to-day situations.

7. TESTING AND EVALUATION

7.1 Testing

Testing is an essential stage in the creation of the vision-voice application, where every module has been tested to perform as expected and deliver a smooth, consistent experience to visually impaired users. Testing comprised a number of stages that consisted of unit testing, integration testing, and user acceptance testing(UAT), each emphasizing different elements of the system.

1. **Unit Testing:** Every module in the app was tested independently to ensure that the fundamental features were functioning as intended. For instance, indoor navigation module was tested for proper QR code scanning and proper room details extraction, while the outdoor navigation module was tested for voice recognition and route generation based on Google Maps. The SOS alert system was also tested to confirm that the GPS coordinates were recorded accurately and that the emergency message and location link were successfully sent.
2. **Integration Testing:** After individual testing, the modules were integrated, and their interactions were tested to ensure seamless communication between them. For instance, the camera feed from the object detection module was integrated with the YOLOv8 model, and the detected objects were relayed to the Text-to-Speech (TTS) engine. Similarly, the barcode scanning module was tested to ensure that the camera, barcode recognition, database retrieval, and TTS feedback worked harmoniously.
3. **User Acceptance Testing (UAT):** User testing was conducted with a group of visually impaired individuals to assess the application's usability and effectiveness in real-world scenarios. Feedback from users was collected to identify any areas of improvement in the interface, voice prompts, and overall ease of use. Based on the feedback, further refinements were made to improve the app's accessibility and user experience.

7.2 Evaluation

The evaluation of the Vision-Voice application focuses on assessing the system's performance, usability, and effectiveness in real-world scenarios. The primary goal of the evaluation was to ensure that the application successfully meets the needs of visually impaired individuals, providing them with reliable navigation, object detection, and emergency assistance.

- **Performance Evaluation:** The system's performance was evaluated by measuring the accuracy and responsiveness of key modules. For the indoor navigation module, the QR code scanning speed and the precision of room detail extraction were tested in various lighting conditions to ensure consistent performance. The outdoor navigation module was assessed for its ability to accurately recognize voice commands and provide real-time directions through Google Maps. The object detection module was evaluated by testing the YOLOv8 model's accuracy in detecting obstacles and providing timely auditory feedback to the user. The system's overall latency, particularly the time taken to process data and deliver Text-to-Speech (TTS) feedback, was closely monitored to ensure a smooth user experience.
- **Usability Testing:** The usability of the application was evaluated by conducting tests with visually impaired individuals. The interface was designed to be simple and accessible, with clear voice prompts guiding users through each step. Feedback from the users was gathered to assess how intuitive and easy the system was to use. Features such as the SOS alert system and barcode scanning module were also tested in real-world scenarios, with users reporting on their effectiveness and ease of access during emergencies and shopping.
- **Effectiveness Assessment:** The evaluation measured how effectively the application fulfilled its purpose. Metrics such as successful object avoidance, the accuracy of product identification in the supermarket, and the time taken to receive emergency help were used to determine the system's overall success in enhancing independence for visually impaired users.

8. USER INTERFACE INTERACTION

8.1 Design Principles

1. Accessibility and User-Centric Design

The primary objective of the system is to ensure that visually impaired users can easily interact with the application without requiring external assistance. The interface is designed with voice-based interaction, eliminating the need for complex touch-based navigation. The use of Text-to-Speech (TTS) technology ensures that users receive clear and comprehensible audio feedback.

2. Real-Time Processing and Efficiency

Since the system involves object detection, navigation, and emergency alert mechanisms, real-time processing is critical. The implementation of YOLOv8 and OpenCV enables quick and accurate object recognition, ensuring that users receive immediate feedback about obstacles. Similarly, Google Maps API integration allows for smooth and responsive outdoor navigation, providing turn-by-turn directions in real time.

3. Modularity and Maintainability

The system is built in a modular manner, with each functionality (indoor navigation, outdoor navigation, SOS alerts, object detection, and product recognition) operating as an independent module. This modular approach allows for ease of updates and enhancements, making it possible to integrate new features without disrupting the existing system.

4. Scalability and Cost-Effectiveness

The Vision-Voice application is designed to be scalable, ensuring compatibility with low-end and high-end smartphones without requiring specialized hardware. By leveraging widely available smartphone cameras and cloud-based processing, the system remains cost-effective and accessible to a larger population.

8.2 Key Features

1. Indoor Navigation via QR Codes

One of the standout features of Vision-Voice is its indoor navigation system that utilizes QR codes placed at various locations such as room doors or hallways. When a visually impaired user scans a QR code, the system retrieves relevant information (such as room number, location description, or instructions) embedded in the code and reads it aloud using Text-to-Speech (TTS). This enables users to independently navigate indoor spaces like malls, offices, hospitals, or schools without needing external help. The feature is easy to implement in most environments and offers a highly effective way to orient visually impaired users.

2. Outdoor Navigation with Voice Assistance

For outdoor navigation, Vision-Voice integrates Google Maps and speech recognition technology, allowing users to input their desired destinations using voice commands. The app generates turn-by-turn navigation instructions and provides continuous guidance through TTS. This feature ensures that users can navigate unfamiliar streets or areas independently, with real-time auditory feedback on their position, direction, and estimated time of arrival. It eliminates the need for manual input, which can be difficult for visually impaired individuals.

3. SOS Alert System

The SOS alert system is a vital feature for ensuring user safety in emergencies. When activated, it sends an emergency message along with the real-time GPS location to pre-defined contacts, providing immediate assistance when needed. This ensures that users are not left stranded in emergency situations and can quickly notify others of their location, offering peace of mind to both users and their loved ones.

4. Real-Time Object Detection

The real-time object detection feature of the app uses the YOLOv8 model to scan the user's surroundings for potential obstacles or objects. The system processes the video feed from the device's camera and identifies objects such as chairs, tables, or people, providing auditory feedback about their location. This helps users avoid collisions and navigate through crowded or cluttered environments, enhancing spatial awareness.

5. Barcode Scanning for Product Identification

Vision-Voice includes a barcode scanning feature to assist users while shopping. By scanning the barcode of products in a supermarket, the app fetches product details from a connected database and reads out the name, price, and other important information. This helps visually impaired individuals make informed purchasing decisions without relying on store attendants, promoting independence in retail environments.

6. Multi-Device Support

The system is designed to be compatible with a wide range of smartphones, making it accessible to a larger population. It utilizes the existing hardware resources (like the smartphone's camera, GPS, and microphone) without requiring specialized or expensive devices, keeping the system cost-effective and widely deployable.

7. Easy User Interface

The user interface (UI) is designed to be simple and intuitive, with minimal interaction required. Since visually impaired users will rely primarily on auditory feedback, the system is built to ensure voice-guided operation. Simple touch gestures or voice commands are all that is needed to activate various features.

10. OUTPUT SCREENS

10.1 Messages App

Fig 10.1.1 TTS Screen

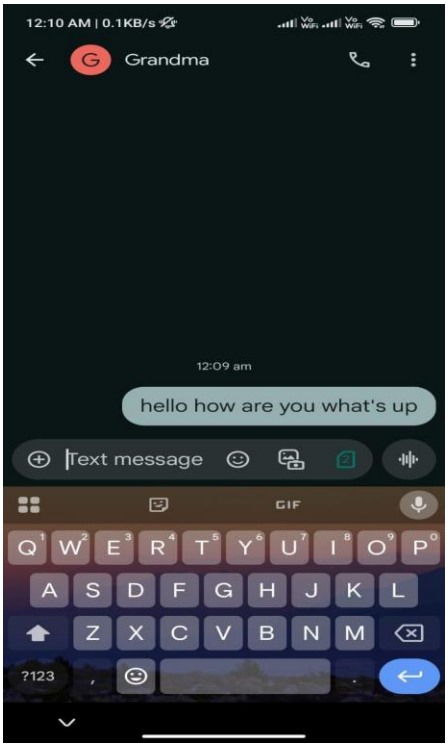

Fig 10.1.2 Message Screen

10.2 Indoor Navigation App

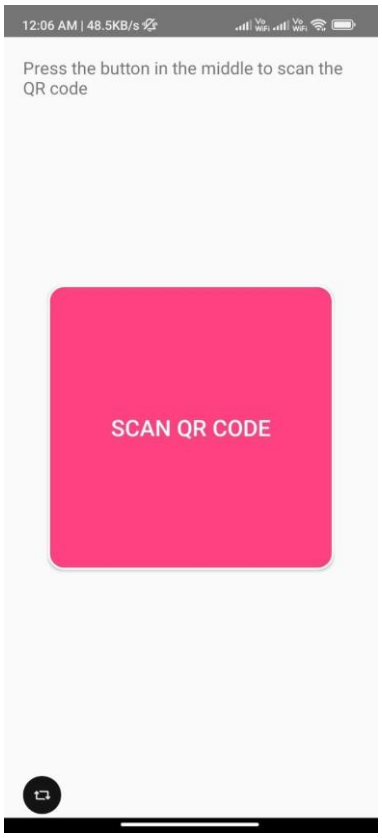

Fig 10.2.1 Home screen.

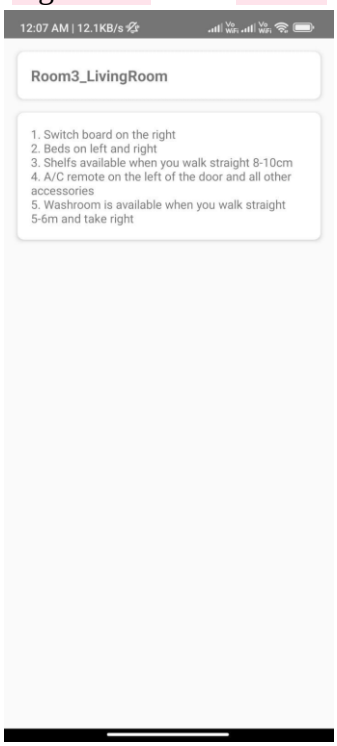

Fig 10.2.2 Room Details Screen

10.3 Outdoor Navigation App

Fig 10.3.1 Home Screen

Fig 10.3.2 TTS Screen

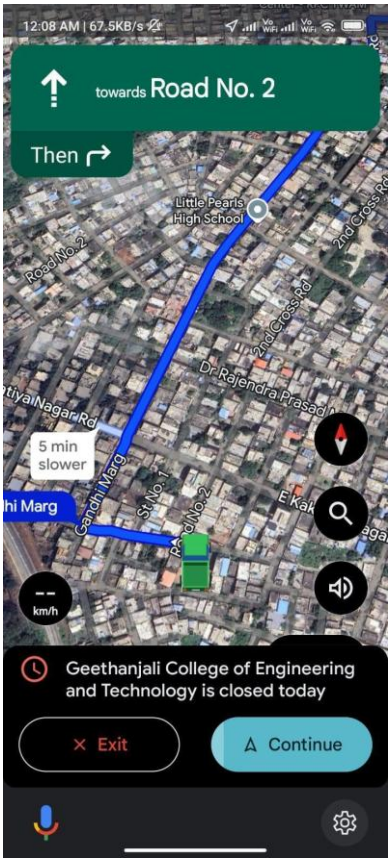

Fig 10.3.3 Navigation Screen

10.4 SOS Alert App

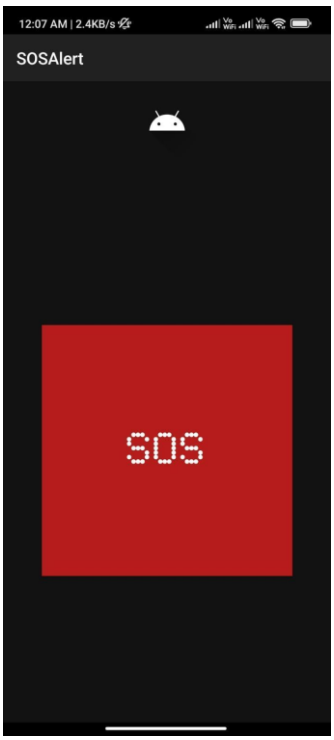

Fig 10.4.1 Home Screen.

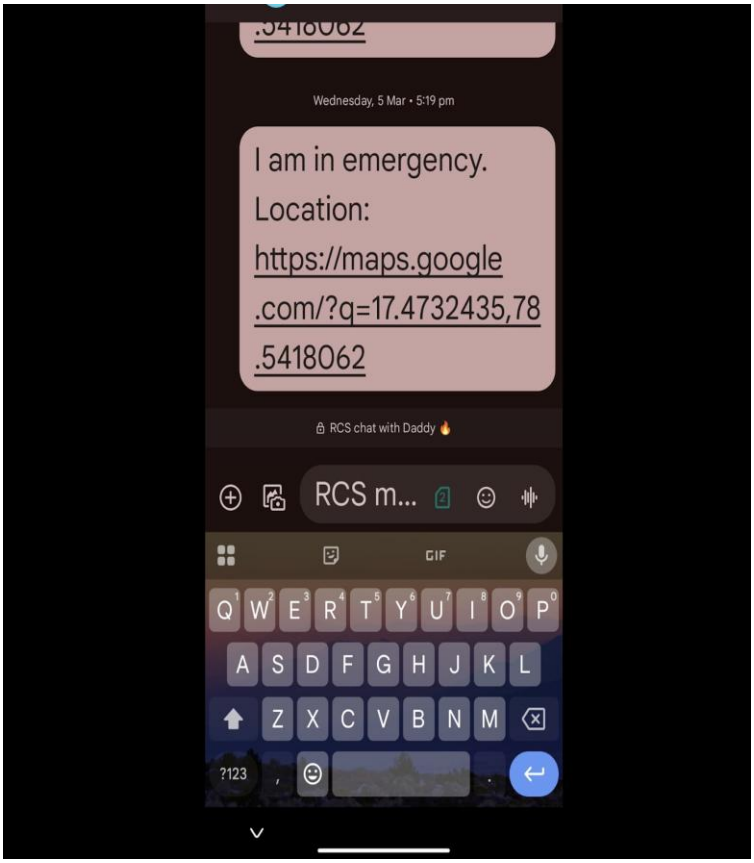

Fig 10.4.2 Emergency message Screen.

10.5 Object Detection

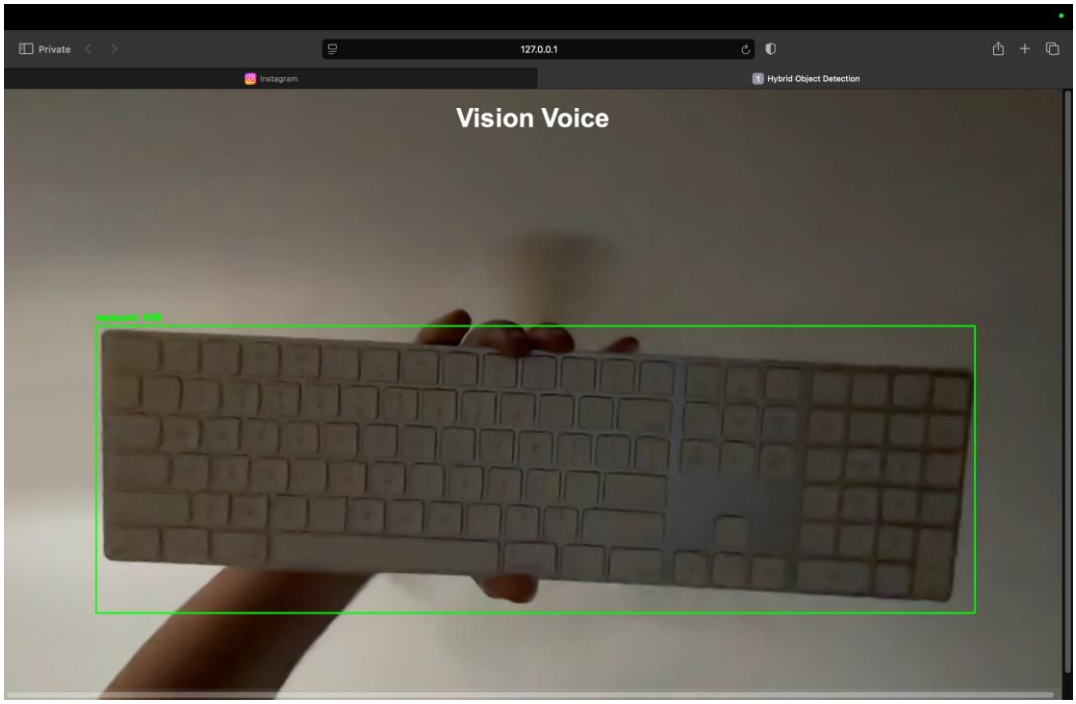

Fig 10.5.1 Object Detection

11. CONCLUSION

The Vision-Voice: Smart Assistant application for Visually Impaired is a contribution to the quality of life and independence of visually impaired individuals. Through the use of innovative and advanced technologies like QR code reading, GPS navigation, text-to-speech technology, and real-time object recognition, the app is solving a series of important and complex issues for visually impaired people. The combination of indoor and outdoor navigation, SOS alerts, object recognition, and product identification in supermarkets and even at home offers a complete and innovative tool for users to navigate around with more confidence and ease.

One of the key strengths of the project is accessibility and ease of use of the application for visually impaired people. The app has been designed with a voice-over feature that allows users to interact with the app manually, without interfering with other operations with some sort of physical activity. It is particularly significant for the visually impaired population, who utilize clear, timely, and accurate audio cues.Moreover, the integration of real-time object detection with YOLO models and OpenCV guarantees users' unrestricted movement and navigation free of obstructions in new surroundings and reduces scope for accidents.Therefore, the SOS alert feature provides users with a quick and effective mode of conveying distress conditions and sending their precise location to loved ones for enhanced safety in outdoor settings.

The success of the deployment of these features highlight how technology can be harnessed to provide pragmatic solutions to everyday problems that visually impaired individuals come across daily. With the use of available technologies like Google Maps, Android development, and TTS systems, the project is guaranteed to be implemented and used without necessarily employing very specialized hardware.

Overall, Vision-Voice is not only a device for helping the blind but also an example of the way innovation and careful design can close the distance between disability and autonomy. Vision-Voice leads the way towards future development, with the prospect of adding other features and rendering the app yet more adaptable, ultimately enhancing the lives of those who use it.

12. FUTURE SCOPE

Vision-Voice: Smart Assistant for Visually Impaired application project holds a lot of promise regarding further development and upgrading in the light of technology's progress and feedback from users through usage. Among the areas of future development will be the detection of objects.As machine learning algorithms such as YOLO advance, better and faster object detection can be implemented into the system.In the future, the object detection ability could detect additional objects, including even smaller or more specific ones, and provide additional detailed information feedback to people, e.g., object measures or real-time warning of hazardous things, enabling one to go even safer navigating in complicated environments.

Another place where things can improve is augmented indoor navigation. Currently, the system has only QR scanning to navigate the user indoors. It should feature Bluetooth beacons or sensor technology for ever more real-time and ongoing indoor way finding independent of QR reading in the future. This enables visually impaired consumers to navigate indoor shopping malls, airports, hospitals, and other indoor complexes independently and more conveniently. Indoor navigation can be dynamically faster and environmentally-adjustable through the use of new rather than obscene technologies such as Wi-Fi signal strength or ultrasonic sensors.

In outdoor navigation, augmented reality and computer vision could combine to enhance the user experience considerably. For example, upcoming versions of the app can provide more intuitive directions through vibrations, along with audio, through neighborhoods and more densely populated areas. In addition, crowdsourced information from other users or city-level data on obstruction, traffic, and construction areas may further enhance route planning and make the application more responsive to actual conditions.

The supermarket guidance feature would also be extended by introducing voice command compatibility with online stores for visually impaired persons to be able to shop independently at home. In short, Vision-Voice can evolve as an even stronger, multi-tasking tool making use of coming technologies like augmented reality, machine learning, and cloud services.

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